

MIT

**Academy of
Engineering**

(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

EDS ASSIGNMENT

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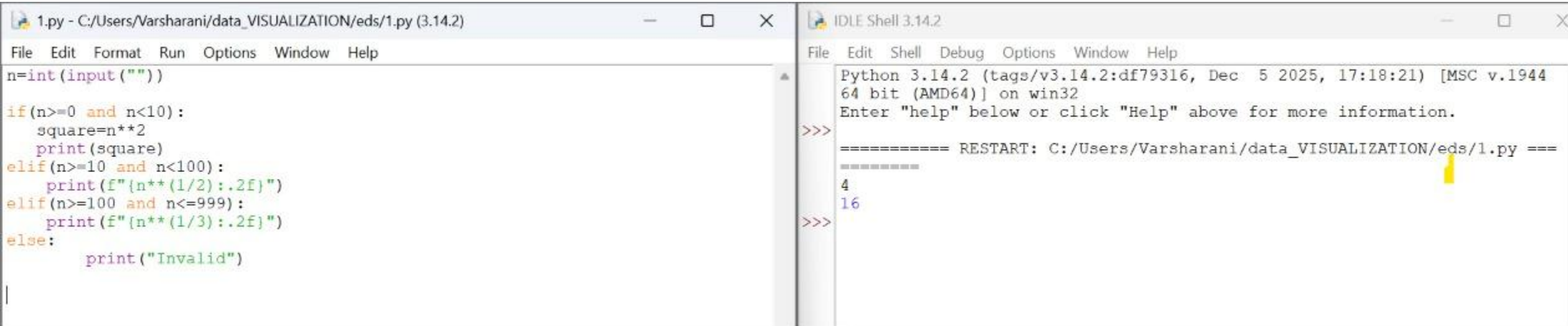
DIV : SE2

BATCH : S24

PRACTICAL NO. 01

Practice Lab Assignments:

1. To accept an object mass in kilograms and velocity in meters per second and display its momentum. Momentum is calculated as $e=mc^2$ where m is the mass of the object and c is its velocity.



The image shows a screenshot of a Python IDE with two windows. The left window, titled '1.py - C:/Users/Varsharani/data_VISUALIZATION/eds/1.py (3.14.2)', contains the following Python code:

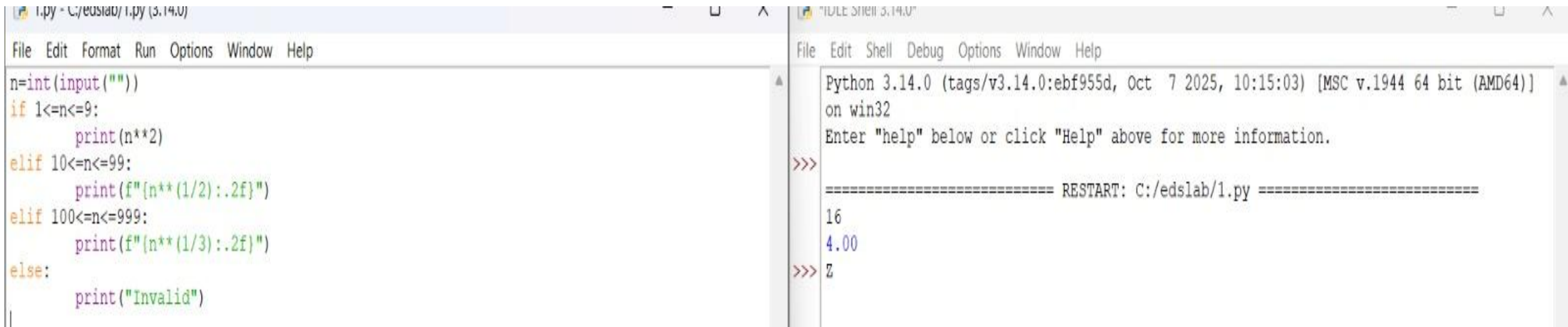
```
n=int(input(""))
if(n>=0 and n<10):
    square=n**2
    print(square)
elif(n>=10 and n<100):
    print(f"{n**(1/2):.2f}")
elif(n>=100 and n<=999):
    print(f"{n**(1/3):.2f}")
else:
    print("Invalid")
```

The right window, titled 'IDLE Shell 3.14.2', shows the execution of the script. It displays the Python version and architecture, followed by a restart message and the output of the script for inputs 4 and 16:

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944
64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Varsharani/data_VISUALIZATION/eds/1.py =====
4
16
>>>
```

2. Write a Python program for following conditions.

- ^ If n is single digit print square of it.
- ^ If n is two digit print squareroot of it.
- ^ If n is three digit print cube root of it.



The image shows a screenshot of a Python IDE with two windows. The left window displays a Python script, and the right window shows the program's execution output.

```
1.py - C:/edslab/1.py (3.14.0)
File Edit Format Run Options Window Help

n=int(input(""))
if 1<=n<=9:
    print(n**2)
elif 10<=n<=99:
    print(f"{n**(1/2):.2f}")
elif 100<=n<=999:
    print(f"{n**(1/3):.2f}")
else:
    print("Invalid")
```

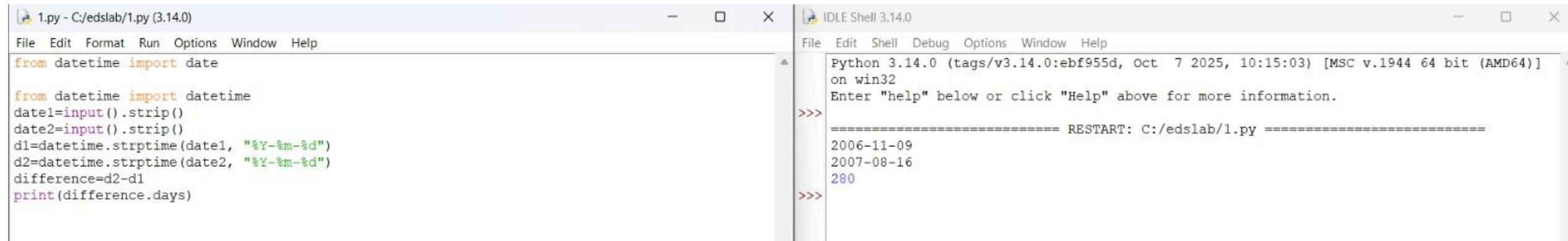
The right window, titled "IDLE Shell 3.14.0", shows the program's execution. It starts with the Python version and system information, followed by a prompt to enter "help". The user enters "16", and the program outputs "4.00". The user then enters "Z", and the program outputs "Invalid".

```
IDLE Shell 3.14.0
File Edit Shell Debug Options Window Help

Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/edslab/1.py =====
16
4.00
>>> Z
Invalid
```

3. Read the birth date and salary in rupees of employees. Perform data transformation for birth date to age and also salary which is in rupees to salary in dollars using functions



The image shows a screenshot of a Python IDE with two windows. The left window, titled '1.py - C:/edslab/1.py (3.14.0)', contains the following Python code:

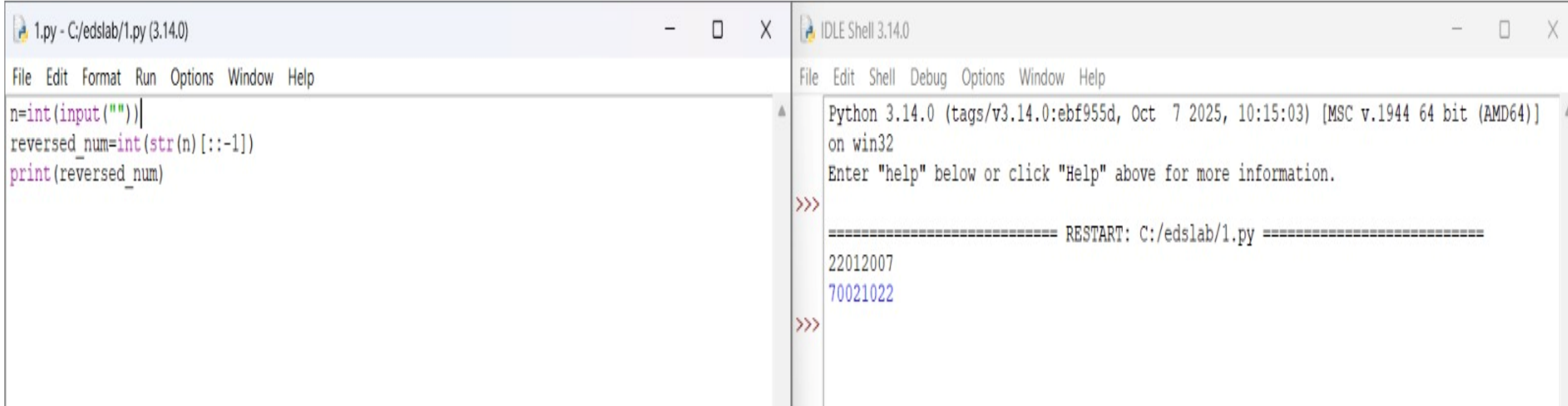
```
from datetime import date

from datetime import datetime
date1=input().strip()
date2=input().strip()
d1=datetime.strptime(date1, "%Y-%m-%d")
d2=datetime.strptime(date2, "%Y-%m-%d")
difference=d2-d1
print(difference.days)
```

The right window, titled 'IDLE Shell 3.14.0', shows the output of the script after execution. It displays the Python version and architecture, followed by a prompt to enter 'help'. The script output shows the dates '2006-11-09' and '2007-08-16', and the calculated age '280'.

```
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/edslab/1.py =====
2006-11-09
2007-08-16
280
>>>
```

4. Print the reverse number of a given number,



The image shows a screenshot of a Python IDE with two windows. The left window, titled '1.py - C:/edslab/1.py (3.14.0)', contains the following Python code:

```
n=int(input(""))
reversed_num=int(str(n)[::-1])
print(reversed_num)
```

The right window, titled 'IDLE Shell 3.14.0', shows the execution output:

```
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/edslab/1.py =====
22012007
70021022
>>>
```

Lab Assignment :

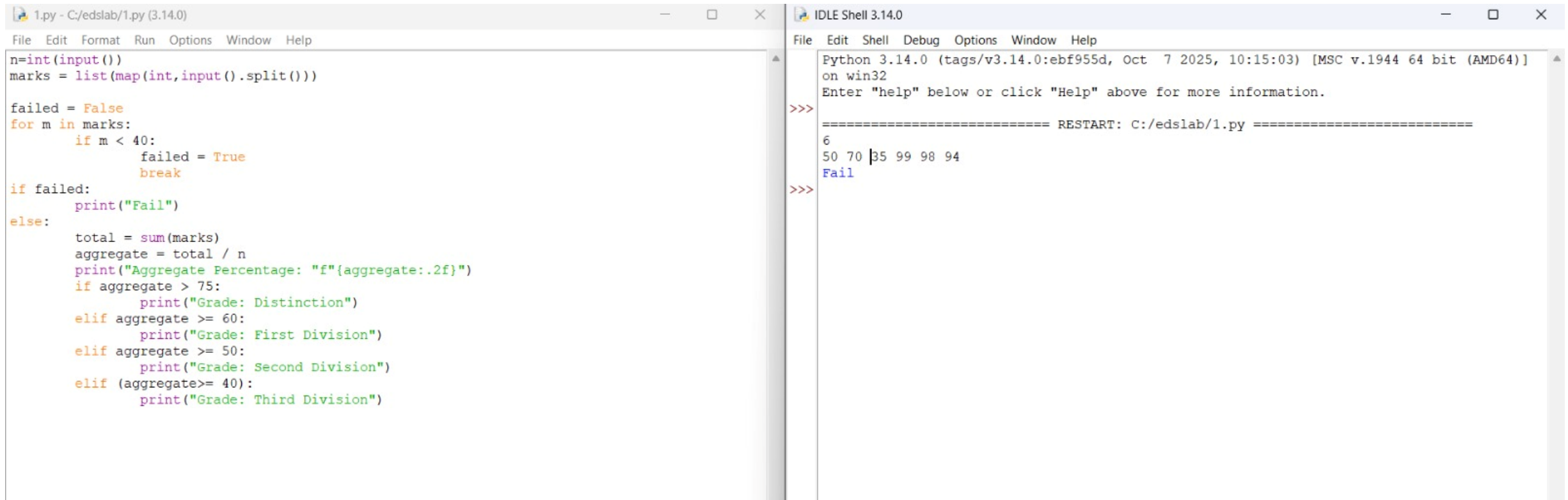
1. To accept students five courses marks and compute his/her result. Student is passing if he/she scores marks equal to and above 40 in each course.

If student scores aggregate greater than 75 percentage , then the grade is distinction.

If aggregate is greater than or equal to 60 and less than 75 then the grade is first division.

If aggregate is greater than or equal 50 and less than 60 , then the grade is second division.

If aggregate is greater than or equal 40 and less than 50 , then the grade is third division.



The image shows a screenshot of a Python IDE with two windows. The left window, titled '1.py - C:/edslab/1.py (3.14.0)', contains the following Python code:

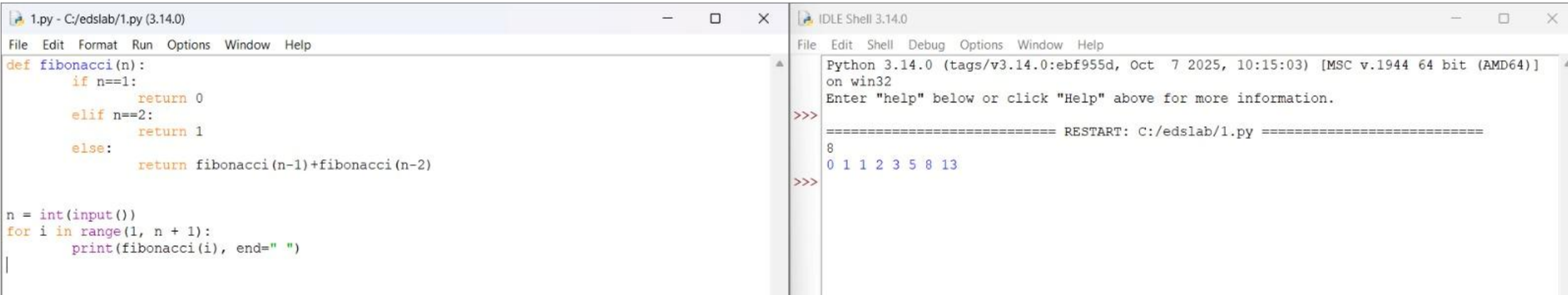
```
n=int(input())
marks = list(map(int,input().split()))

failed = False
for m in marks:
    if m < 40:
        failed = True
        break
if failed:
    print("Fail")
else:
    total = sum(marks)
    aggregate = total / n
    print("Aggregate Percentage: "f"{aggregate:.2f}")
    if aggregate > 75:
        print("Grade: Distinction")
    elif aggregate >= 60:
        print("Grade: First Division")
    elif aggregate >= 50:
        print("Grade: Second Division")
    elif (aggregate>= 40):
        print("Grade: Third Division")
```

The right window, titled 'IDLE Shell 3.14.0', shows the execution of the script. It displays the Python version and architecture, followed by a prompt to enter 'help' for more information. The user has entered '6' for the number of courses and '50 70 35 99 98 94' for the marks. The script has executed and printed 'Fail'.

```
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/edslab/1.py =====
6
50 70 35 99 98 94
Fail
>>>
```

2. Solve the Fibonacci sequence using recursive function in Python.



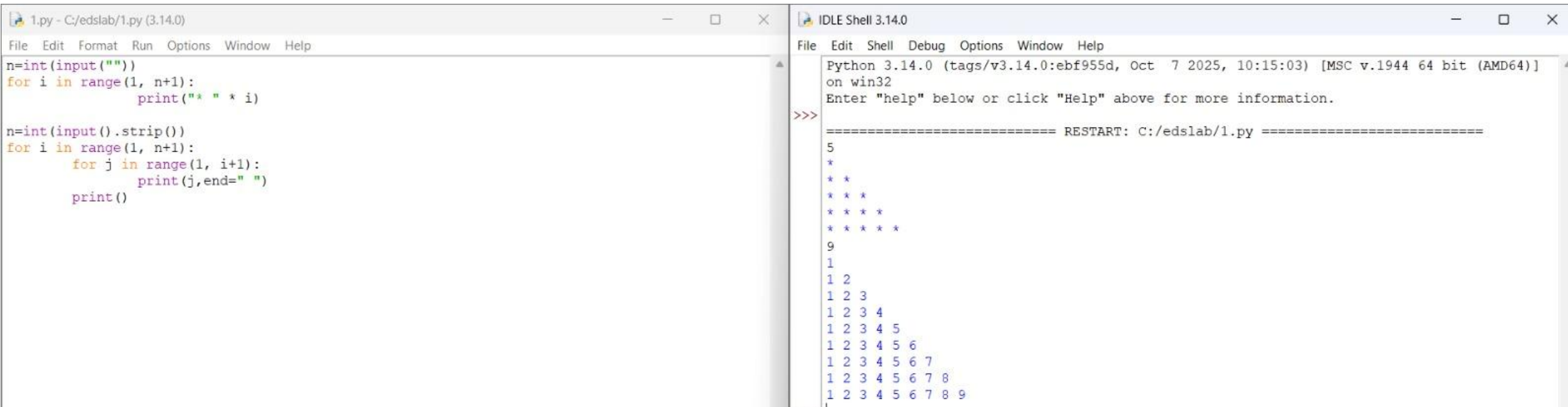
The image shows a Python IDE with two windows. The left window, titled '1.py - C:/edslab/1.py (3.14.0)', contains the following code:

```
def fibonacci(n):  
    if n==1:  
        return 0  
    elif n==2:  
        return 1  
    else:  
        return fibonacci(n-1)+fibonacci(n-2)  
  
n = int(input())  
for i in range(1, n + 1):  
    print(fibonacci(i), end=" ")  
|
```

The right window, titled 'IDLE Shell 3.14.0', shows the execution of the program. It displays the Python version and architecture, followed by a prompt to enter 'help'. The program then restarts, and the output of the Fibonacci sequence for the first 8 terms is displayed:

```
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)]  
on win32  
Enter "help" below or click "Help" above for more information.  
>>>  
===== RESTART: C:/edslab/1.py =====  
8  
0 1 1 2 3 5 8 13  
>>>
```

3 . Write a Python program to print different patterns



The image shows a screenshot of a Python IDE with two windows. The left window, titled '1.py - C:/edslab/1.py (3.14.0)', contains the following Python code:

```
n=int(input(""))
for i in range(1, n+1):
    print("* " * i)

n=int(input().strip())
for i in range(1, n+1):
    for j in range(1, i+1):
        print(j,end=" ")
    print()
```

The right window, titled 'IDLE Shell 3.14.0', shows the output of the program. It displays the Python version and system information, followed by a prompt for user input. The output shows two patterns: a star pattern and a number pattern.

```
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/edslab/1.py =====
5
*
* *
* * *
* * * *
* * * * *
9
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
1 2 3 4 5 6
1 2 3 4 5 6 7
1 2 3 4 5 6 7 8
1 2 3 4 5 6 7 8 9
```


PRACTICALNO.02

Practice Lab Assignment :

1.Perform all List Operations, Dictionary Operations.

```
1.py - C:/edslab/1.py (3.14.0)
File Edit Format Run Options Window Help
list=[]
while True:
    print("1. Add")
    print("2. Remove")
    print("3. Display")
    print("4. Quit")
    n = int(input("Enter choice: "))
    if n == 1:
        add = int(input("Integer: "))
        list.append(add)
        print(f"List after adding: {list}")
    elif n == 2:
        if len(list) == 0:
            print("List is empty")
        elif len(list) != 0:
            remove=int(input("Integer: "))
            if remove not in list :
                print("Element not found")
            else:
                list.remove(remove)
                print(f"List after removing: {list}")
    elif n == 3:
        if len(list) == 0:
            print("List is empty")
        else:
            print(list)
    elif n == 4:
        break
    else:
        print("Invalid choice")
# Initial dictionary

IDLE Shell 3.14.0
File Edit Shell Debug Options Window Help
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 1
Integer: 299
List after adding: [299]
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 299
Invalid choice
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 3
[299]
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 2
Integer: 299
List after removing: []
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 4
```

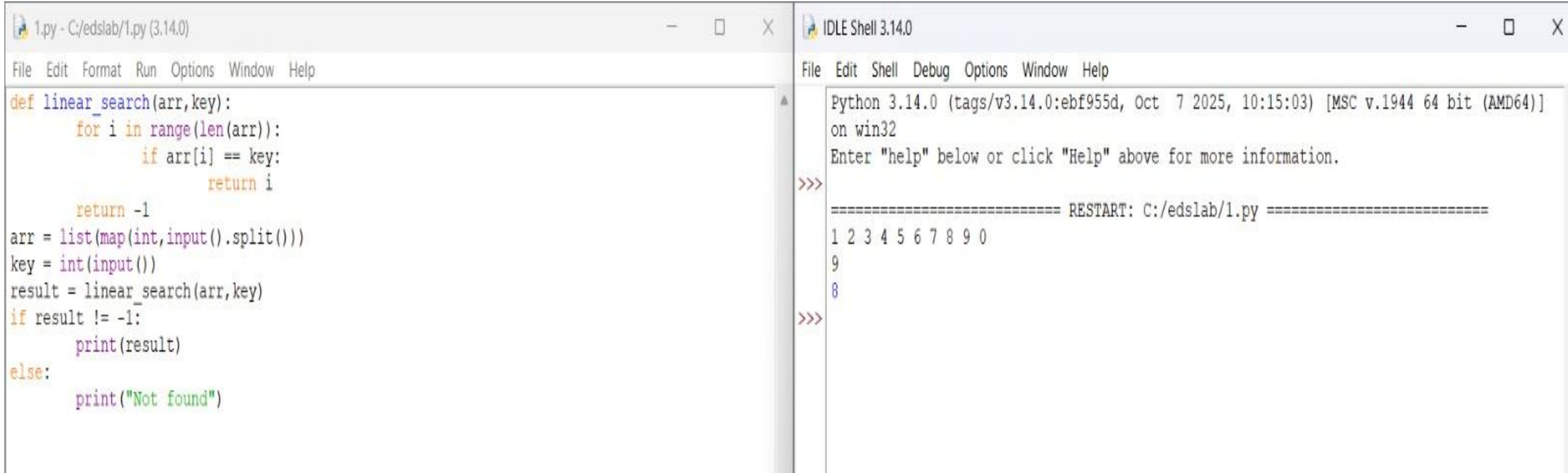
2.Perform all Dictionary Operations

```
# Initial dictionary
student = {
    1: "Amit",
    2: "Riya",
    3: "Kiran",
    4: "Neha",
    5: "Arjun",
    6: "Pooja",
    7: "Rahul",
    8: "Sneha",
    9: "Vikram",
    10: "Anjali"
}
print("Original Dictionary:", student)
key1 = int(input())
value1 = input()
student.update({key1: value1})
print("After Insertion:", student)
key2 = int(input())
value2 = input()
if key2 in student.keys():
    student.update({key2: value2})
print("After Update:", student)
key3 = int(input())
if key3 in student.keys():
    student.pop(key3)
print("After Deletion:", student)
print("Traversing Dictionary:")
for key, value in student.items():
    print(f"{key} : {value}")
```

```
2. Remove
3. Display
4. Quit
Enter choice: 4
Original Dictionary: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali'}
11
varsharani
After Insertion: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 11: 'varsharani'}
8
pranjali
After Update: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'pranjali', 9: 'Vikram', 10: 'Anjali', 11: 'varsharani'}
2
After Deletion: {1: 'Amit', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'pranjali', 9: 'Vikram', 10: 'Anjali', 11: 'varsharani'}
Traversing Dictionary:
1 : Amit
3 : Kiran
4 : Neha
5 : Arjun
6 : Pooja
7 : Rahul
8 : pranjali
9 : Vikram
10 : Anjali
11 : varsharani
>>> 1
1
```

Lab Assignment :

1. Select the number from the entered list and find its position in Python (use Linear Search)



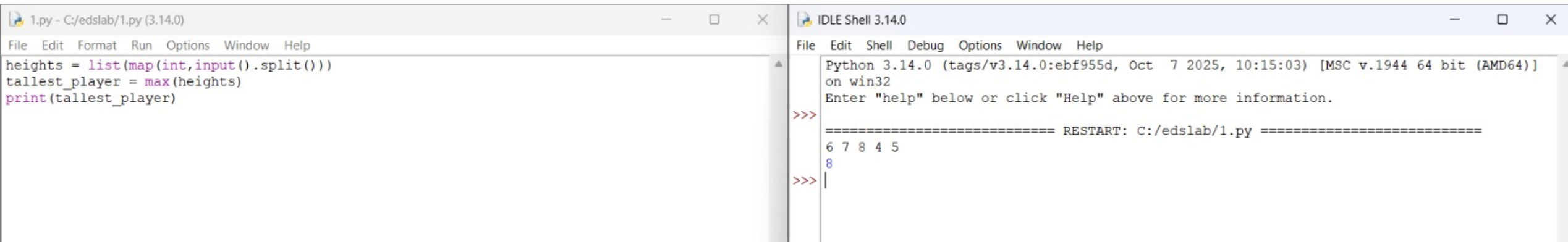
The image shows a screenshot of a Python IDE with two windows. The left window, titled '1.py - C:/edslab/1.py (3.14.0)', contains the following Python code:

```
def linear_search(arr, key):  
    for i in range(len(arr)):  
        if arr[i] == key:  
            return i  
    return -1  
  
arr = list(map(int, input().split()))  
key = int(input())  
result = linear_search(arr, key)  
if result != -1:  
    print(result)  
else:  
    print("Not found")
```

The right window, titled 'IDLE Shell 3.14.0', shows the execution of the program. It displays the Python version and architecture, followed by a prompt to enter 'help' for more information. The user has entered '9' and '8' as input, and the program has restarted. The output shows the list '1 2 3 4 5 6 7 8 9 0' and the position '9' for the key '8'.

```
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)]  
on win32  
Enter "help" below or click "Help" above for more information.  
>>>  
===== RESTART: C:/edslab/1.py =====  
1 2 3 4 5 6 7 8 9 0  
9  
8  
>>>
```

2. Choose cricket team of eleven players find the captain of the team (consider tallest person as a captain) using dictionary.



The image shows a Python IDE with two windows. The left window, titled '1.py - C:/edslab/1.py (3.14.0)', contains the following code:

```
heights = list(map(int, input().split()))
tallest_player = max(heights)
print(tallest_player)
```

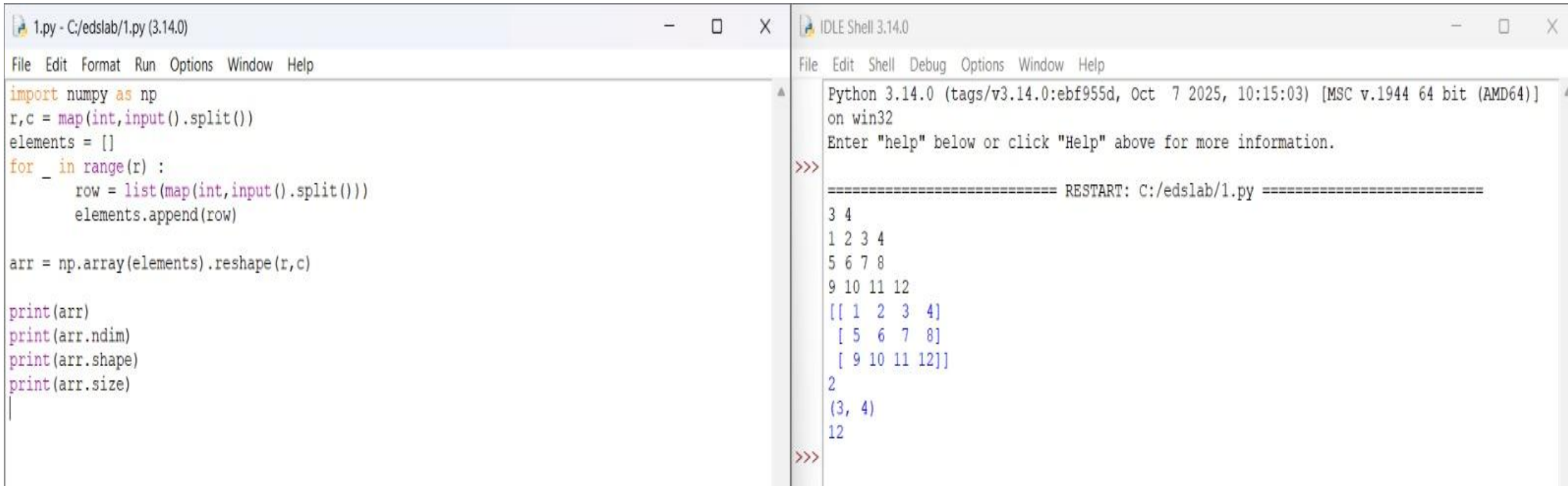
The right window, titled 'IDLE Shell 3.14.0', shows the execution of the script. It displays the Python version and architecture, followed by a prompt to enter 'help' for more information. The script has been restarted, and the input '6 7 8 4 5' has been entered. The output '8' is displayed, indicating the tallest player's height.

```
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/edslab/1.py =====
6 7 8 4 5
8
>>>
```

PRACTICAL NO. 03

Practice Lab Assignment:

1. Perform all the NumPy operations in Python.



The image shows a screenshot of a Python IDE with two windows. The left window, titled '1.py - C:/edslab/1.py (3.14.0)', contains a Python script that reads input from the user, processes it into a NumPy array, and prints its properties. The right window, titled 'IDLE Shell 3.14.0', shows the execution output of the script.

```
File Edit Format Run Options Window Help
import numpy as np
r,c = map(int,input().split())
elements = []
for _ in range(r) :
    row = list(map(int,input().split()))
    elements.append(row)

arr = np.array(elements).reshape(r,c)

print(arr)
print(arr.ndim)
print(arr.shape)
print(arr.size)
```

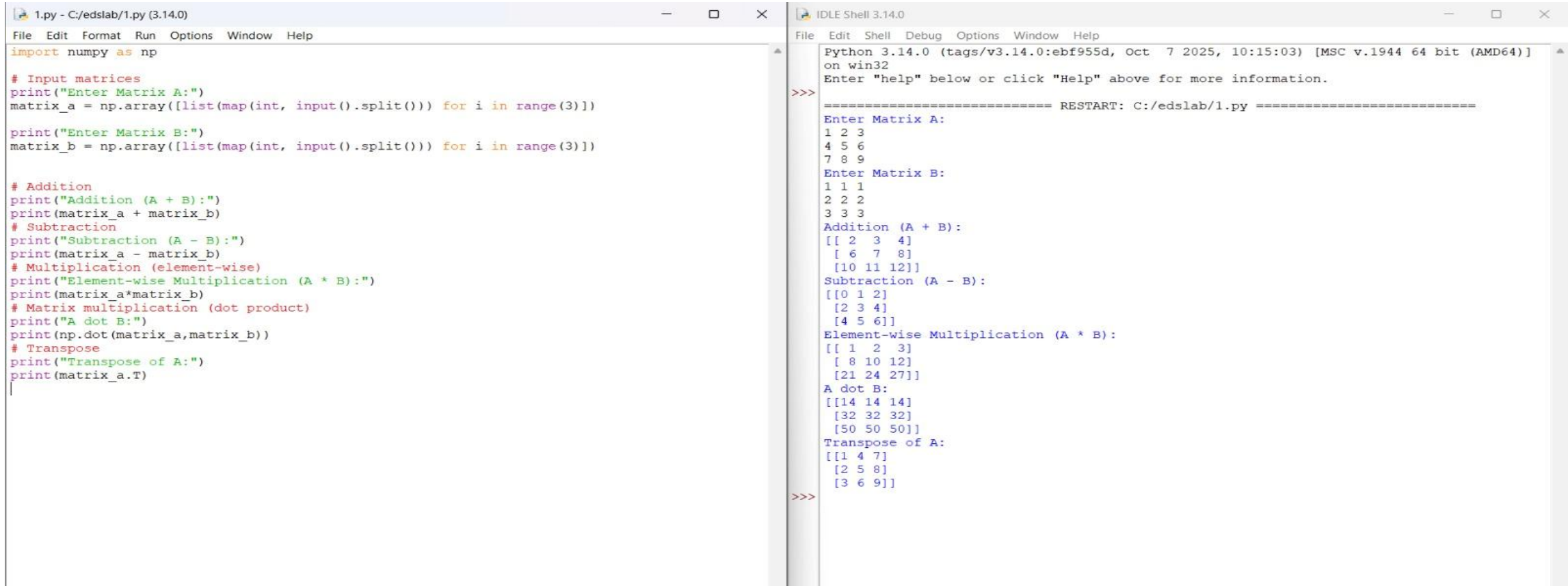
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.

```
>>>
===== RESTART: C:/edslab/1.py =====
3 4
1 2 3 4
5 6 7 8
9 10 11 12
[[ 1  2  3  4]
 [ 5  6  7  8]
 [ 9 10 11 12]]
2
(3, 4)
12
>>>
```

Lab Assignment:

Prepare/Take dataset for any real life application. Read a dataset into an array. Perform following operations on it as:

- Perform all matrix operations
- Horizontal and vertical stacking of NumPy Arrays



The image shows a Python IDE with two windows. The left window, titled '1.py - C:/edslab/1.py (3.14.0)', contains a script for matrix operations. The right window, titled 'IDLE Shell 3.14.0', shows the execution output of the script.

```
1.py - C:/edslab/1.py (3.14.0)
File Edit Format Run Options Window Help
import numpy as np

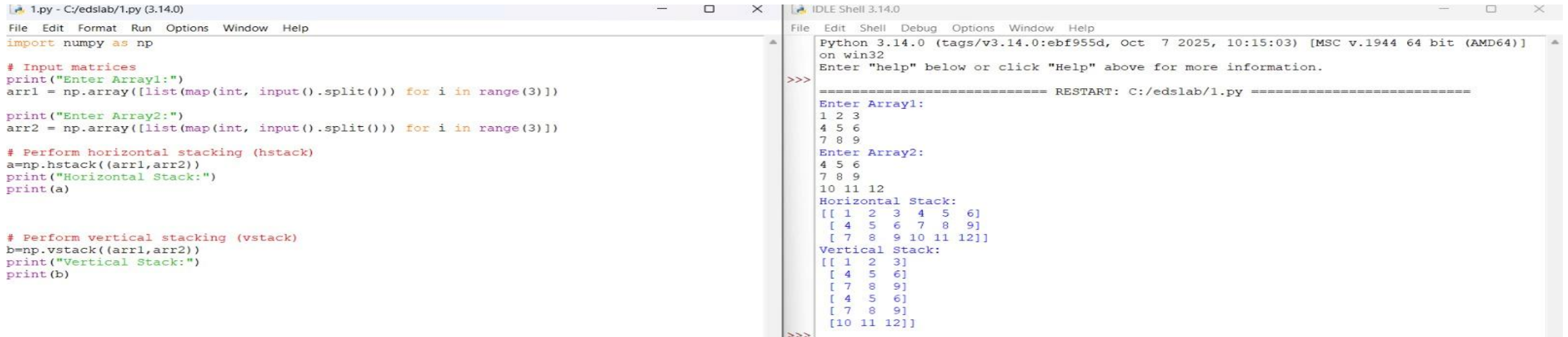
# Input matrices
print("Enter Matrix A:")
matrix_a = np.array([list(map(int, input().split())) for i in range(3)])

print("Enter Matrix B:")
matrix_b = np.array([list(map(int, input().split())) for i in range(3)])

# Addition
print("Addition (A + B):")
print(matrix_a + matrix_b)
# Subtraction
print("Subtraction (A - B):")
print(matrix_a - matrix_b)
# Multiplication (element-wise)
print("Element-wise Multiplication (A * B):")
print(matrix_a*matrix_b)
# Matrix multiplication (dot product)
print("A dot B:")
print(np.dot(matrix_a,matrix_b))
# Transpose
print("Transpose of A:")
print(matrix_a.T)

IDLE Shell 3.14.0
File Edit Shell Debug Options Window Help
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/edslab/1.py =====
Enter Matrix A:
1 2 3
4 5 6
7 8 9
Enter Matrix B:
1 1 1
2 2 2
3 3 3
Addition (A + B):
[[ 2  3  4]
 [ 6  7  8]
 [10 11 12]]
Subtraction (A - B):
[[ 0  1  2]
 [ 2  3  4]
 [ 4  5  6]]
Element-wise Multiplication (A * B):
[[ 1  2  3]
 [ 8 10 12]
 [21 24 27]]
A dot B:
[[14 14 14]
 [32 32 32]
 [50 50 50]]
Transpose of A:
[[1 4 7]
 [2 5 8]
 [3 6 9]]
>>>
```


- Custom sequence generation



The screenshot displays a Python IDE with two windows. The left window, titled '1.py - C:/edslab/1.py (3.14.0)', contains the following code:

```
import numpy as np

# Input matrices
print("Enter Array1:")
arr1 = np.array([list(map(int, input().split())) for i in range(3)])

print("Enter Array2:")
arr2 = np.array([list(map(int, input().split())) for i in range(3)])

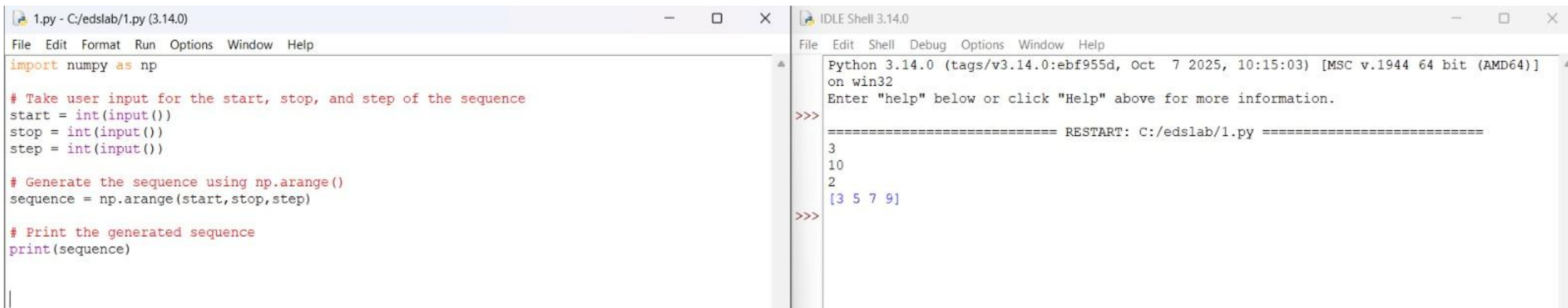
# Perform horizontal stacking (hstack)
a=np.hstack((arr1,arr2))
print("Horizontal Stack:")
print(a)

# Perform vertical stacking (vstack)
b=np.vstack((arr1,arr2))
print("Vertical Stack:")
print(b)
```

The right window, titled 'IDLE Shell 3.14.0', shows the execution output:

```
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/edslab/1.py =====
Enter Array1:
1 2 3
4 5 6
7 8 9
Enter Array2:
4 5 6
7 8 9
10 11 12
Horizontal Stack:
[[ 1  2  3  4  5  6]
 [ 4  5  6  7  8  9]
 [ 7  8  9 10 11 12]]
Vertical Stack:
[[ 1  2  3]
 [ 4  5  6]
 [ 7  8  9]
 [ 4  5  6]
 [ 7  8  9]
 [10 11 12]]
>>>
```

- Arithmetic and Statistical Operations



The screenshot displays a Python IDE with two windows. The left window, titled '1.py - C:/edslab/1.py (3.14.0)', contains the following code:

```
import numpy as np

# Take user input for the start, stop, and step of the sequence
start = int(input())
stop = int(input())
step = int(input())

# Generate the sequence using np.arange()
sequence = np.arange(start,stop,step)

# Print the generated sequence
print(sequence)
```

The right window, titled 'IDLE Shell 3.14.0', shows the execution output:

```
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/edslab/1.py =====
3
10
2
[3 5 7 9]
>>>
```

• Bitwise Operators

```
1.py - C:/edslab/1.py (3.14.0)
File Edit Format Run Options Window Help
import numpy as np
def array_operations(A, B):
    # Convert A and B to NumPy arrays
    A = np.array(A)
    B = np.array(B)
    # Arithmetic Operations
    sum_result = A+B
    diff_result = A-B
    prod_result = A*B

    # Statistical Operations
    mean_A = np.mean(A)
    median_A = np.median(A)
    std_dev_A = np.std(A)

    # Bitwise Operations
    and_result = np.bitwise_and(A,B)
    or_result = np.bitwise_or(A,B)
    xor_result = np.bitwise_xor(A,B)

    # Output results with one space between each element
    print("Element-wise Sum:", ' '.join(map(str, sum_result)))
    print("Element-wise Difference:", ' '.join(map(str, diff_result)))
    print("Element-wise Product:", ' '.join(map(str, prod_result)))

    print(f"Mean of A: {mean_A}")
    print(f"Median of A: {median_A}")
    print(f"Standard Deviation of A: {std_dev_A}")

    print("Bitwise AND:", ' '.join(map(str, and_result)))
    print("Bitwise OR:", ' '.join(map(str, or_result)))
    print("Bitwise XOR:", ' '.join(map(str, xor_result)))

A = list(map(int, input().split())) # Elements of array A
B = list(map(int, input().split())) # Elements of array B
array_operations(A, B)

|
```

```
IDLE Shell 3.14.0
File Edit Shell Debug Options Window Help
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/edslab/1.py =====
1 2 3 4
5 6 7 8
Element-wise Sum: 6 8 10 12
Element-wise Difference: -4 -4 -4 -4
Element-wise Product: 5 12 21 32
Mean of A: 2.5
Median of A: 2.5
Standard Deviation of A: 1.118033988749895
Bitwise AND: 1 2 3 0
Bitwise OR: 5 6 7 12
Bitwise XOR: 4 4 4 12
>>>
```

• Copying and viewing arrays

```
1.py - C:/edslab/1.py (3.14.0)
File Edit Format Run Options Window Help
import numpy as np

inputlist = list(map(int, input().split(" ")))

# Original array
original_array = np.array(inputlist)

# Create a view
view_array = original_array.view()

# Create a copy
copy_array = original_array.copy()

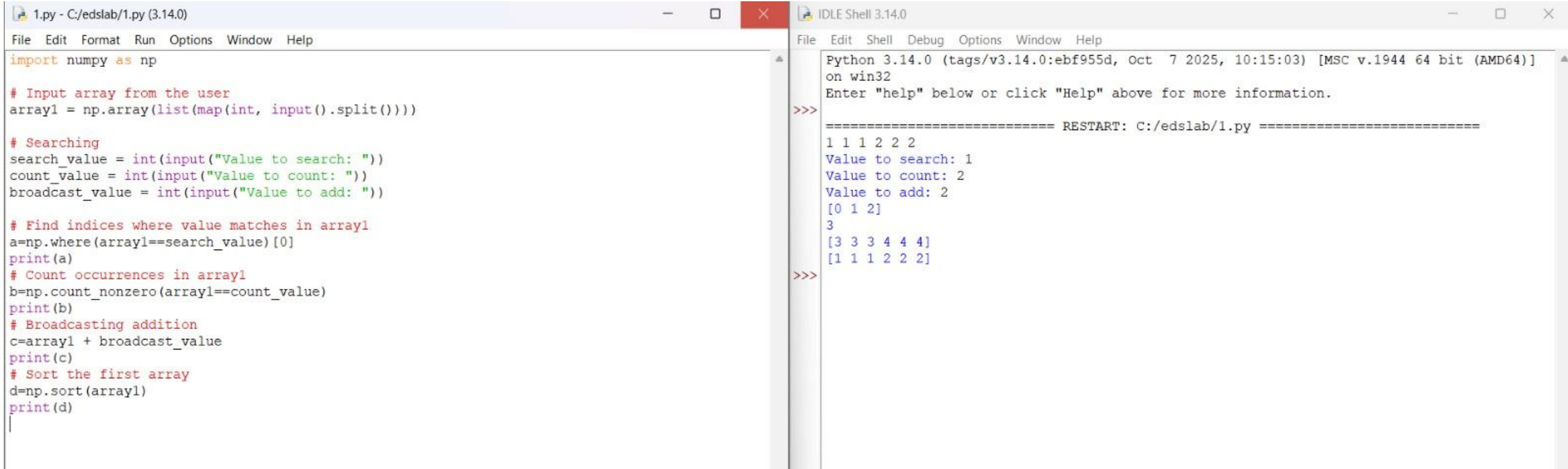
# Modify the view
view_array[0] = 99
print("Original array after modifying view:", original_array)
print("View array:", view_array)

# Modify the copy
copy_array[1] = 88
print("Original array after modifying copy:", original_array)
print("Copy array:", copy_array)
```

```
IDLE Shell 3.14.0
File Edit Shell Debug Options Window Help
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/edslab/1.py =====
10 20 30 40 50 60 70
Original array after modifying view: [99 20 30 40 50 60 70]
View array: [99 20 30 40 50 60 70]
Original array after modifying copy: [99 20 30 40 50 60 70]
Copy array: [10 88 30 40 50 60 70]
>>>
```


- Data Stacking
- Searching
- Sorting
- Broadcasting
- Counting



The image shows a Python IDE with two windows. The left window, titled '1.py - C:/edslab/1.py (3.14.0)', contains the following code:

```
import numpy as np

# Input array from the user
array1 = np.array(list(map(int, input().split()))))

# Searching
search_value = int(input("Value to search: "))
count_value = int(input("Value to count: "))
broadcast_value = int(input("Value to add: "))

# Find indices where value matches in array1
a=np.where(array1==search_value)[0]
print(a)
# Count occurrences in array1
b=np.count_nonzero(array1==count_value)
print(b)
# Broadcasting addition
c=array1 + broadcast_value
print(c)
# Sort the first array
d=np.sort(array1)
print(d)
```

The right window, titled 'IDLE Shell 3.14.0', shows the execution output:

```
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/edslab/1.py =====
1 1 1 2 2 2
Value to search: 1
Value to count: 2
Value to add: 2
[0 1 2]
3
[3 3 3 4 4 4]
[1 1 1 2 2 2]
>>>
```

PRACTICAL NO. 04

Practice Lab Assignment:

1. Perform all the pandas operations in Python.

```
import pandas as pd

# Take inputs from the user to create a list of numbers
numbers = list(map(int, input().split()))

# Create a Pandas series from the list of numbers
series = pd.Series(numbers)

# Grouping by even and odd numbers and calculating the mean
grouped = series.groupby(series % 2 == 0).mean()

# Display the mean of even and odd numbers with labels
grouped.index = ['Even' if is_even else 'Odd' for is_even in grouped.index]
print("Mean of even and odd numbers:")
print(grouped)
```

Average time

0.009 s

8.67 ms



Maximum time

0.022 s

22.00 ms



✓ 3 out of 3 shown test case(s) p

✓ 3 out of 3 hidden test case(s) p

✓ Test case 1 22 ms

Debug



Expected output

1 2 3 4 5 6 7 8 9 10

Mean of even and odd numbers:

Odd 5.0

Even 6.0

dtype: float64

Actual output

1 2 3 4 5 6 7 8 9 10

Mean of even and odd numbers:

Odd 5.0

Even 6.0

dtype: float64

✓ Test case 2 8 ms



✓ Test case 3 7 ms



Lab Assignment:

Read any real life dataset. Store the data into Data Frames. Identify 10 grains for the given dataset. Implement all 20 grains using Pandas methods.

```
import pandas as pd

# Provided dictionary of lists
data = {
    'Name': ['Alice', 'Bob', 'Charlie'],
    'Age': [25, 30, 35],
}

# Convert the dictionary to a DataFrame
df = pd.DataFrame(data)

# Display the original DataFrame
print("Original DataFrame:")
print(df)

# Adding a new row
Name=input("New name: ")
Age=int(input("New age: "))
new={'Name':Name, 'Age':Age}
df=df.append(new,ignore_index=True)

# Display the DataFrame after adding a new row
print("After adding a row:\n",df)

# Modifying a row
index=int(input("Index of row to modify: "))
age=int(input("New age: "))
df.at[index,"Age"]=age
```

```
# Deleting a row
delt=int(input("Index of row to delete: "))
df=df.drop(delt).reset_index(drop=True)
# Display the DataFrame after deleting a row
print("After deleting a row:")
print(df)

# Adding a new column
genders = input("Enter genders separated by space: ").split()
df["Gender"] = genders

# Display the DataFrame after adding a new column
print("After adding a new column:")
print(df)

# Modifying a column
df['Name']=df['Name'].str.upper()
# Display the DataFrame after modifying a column
print("After modifying a column:")
print(df)

# Deleting a column
df=df.drop(columns=['Age'])
# Display the DataFrame after deleting a column
print("After deleting a column:")
print(df)
```

Average time

0.067 s

67.00 ms



Maximum time

0.071 s

71.00 ms



✓ 1 out of 1 shown test case(s) passed

✓ 1 out of 1 hidden test case(s) passed

New age: 27

After modifying a row:

.....Name..Age

0...Alice...27

1.....Bob...30

2..Charlie...35

3...Susan...29

Index of row to delete: 3

After deleting a row:

.....Name..Age

0...Alice...27

1.....Bob...30

2..Charlie...35

Enter genders separated by space: F M M

After adding a new column:

.....Name..Age..Gender

0...Alice...27.....F

1.....Bob...30.....M

2..Charlie...35.....M

After modifying a column:

.....Name..Age..Gender

0...ALICE...27.....F

1.....BOB...30.....M

2..CHARLIE...35.....M

After deleting a column:

.....Name..Gender

New age: 27

After modifying a row:

.....Name..Age

0...Alice...27

1.....Bob...30

2..Charlie...35

3...Susan...29

Index of row to delete: 3

After deleting a row:

.....Name..Age

0...Alice...27

1.....Bob...30

2..Charlie...35

Enter genders separated by space: F M M

After adding a new column:

.....Name..Age..Gender

0...Alice...27.....F

1.....Bob...30.....M

2..Charlie...35.....M

After modifying a column:

.....Name..Age..Gender

0...ALICE...27.....F

1.....BOB...30.....M

2..CHARLIE...35.....M

After deleting a column:

.....Name..Gender

The Sample Grains for Sales Dataset as:

Which was the best month for sales? How much was earned that month?

```
import pandas as pd

# Prompt the user for the file name
file_name = input()

# Load the data
df = pd.read_csv(file_name)
df['Date']=pd.to_datetime(df['Date'])
df['Month']=df['Date'].dt.to_period('M').astype(str)

# Find the month with the highest total sales
df['Sales']=df['Quantity']*df["Price"]
monthly_sales=df.groupby('Month')['Sales'].sum()
best_month =monthly_sales.idxmax()
highest_sales = monthly_sales.max()

print(f"Best month: {best_month}")
print(f"Total sales: ${highest_sales:.2f}")
```

Average time 0.072 s 71.67 ms	Maximum time 0.140 s 140.00 ms	✓ 1 out of 1 shown test case(s) passed ✓ 2 out of 2 hidden test case(s) passed
--	---	---

✓ Test case 1 140 ms Debug

Expected output	Actual output
sales_data.csv	sales_data.csv
Best month: 2025-01	Best month: 2025-01
Total sales: \$1210.00	Total sales: \$1210.00

Which product sold the most? Why do you think it did?

```
import pandas as pd

# Prompt the user for the file name
file_name = input()

# Load the data
df = pd.read_csv(file_name)
product_sales=df.groupby('Product')
['Quantity'].sum().reset_index()

# Find the product with the highest total quantity sold
best_product
=product_sales.loc[product_sales['Quantity'].idxmax
(),'Product']
highest_quantity = product_sales['Quantity'].max()

# Display the result
print(f"Best selling product: {best_product}")
print(f"Total quantity sold: {highest_quantity}")
```

sales_data.csv

Best selling product: Product A

Total quantity sold: 23

Which city sold the most products?

```
import pandas as pd

# Prompt the user for the file name
file_name = input()

# Load the data
df = pd.read_csv(file_name)

city_wise_sales=df.groupby('City')['Quantity'].sum()

# write the code..
best_city= city_wise_sales.idxmax()
# Display the result
print(f"City sold the most products: {best_city}")
```

sales_data.csv

City sold the most products: Los Angeles ↵

What Products are most often sold together?

```
import pandas as pd
from itertools import combinations
from collections import Counter

# Prompt user to input the file name
file_name = input()

# Read data from the specified CSV file
df = pd.read_csv(file_name)
grouped = df.groupby("Date")["Product"].apply(list)
pairs_counter = Counter()
for products in grouped:
    → pairs = combinations(sorted(products), 2)
    → pairs_counter.update(pairs)
max_frequency = max(pairs_counter.values())
most_common_pairs = [(pair, freq) for pair, freq in
pairs_counter.items() if freq == max_frequency]
for (product1, product2), frequency in most_common_pairs:
    → print(f"{product1} and {product2}: {frequency} times")
... ..
```

sales_data.csv

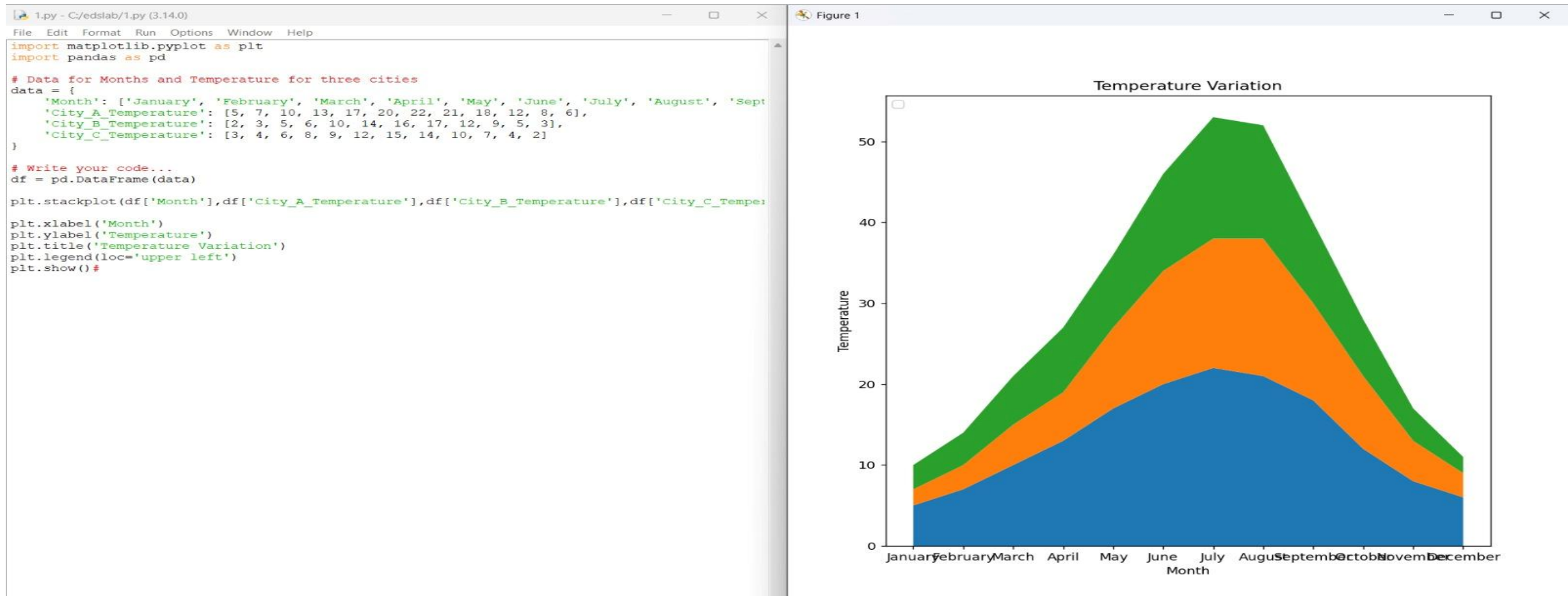
Product A and Product B: 2 times ↵

Product A and Product C: 2 times ↵

PRACTICAL NO. 05

Practice Lab Assignment:

Install Matplotlib library. Draw basic graphs for sales dataset using Matplotlib.



Lab Assignment:

For Titanic dataset, Perform data analysis. Identify 5 grains for a given dataset.

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset from the CSV file
df = pd.read_csv('titanic.csv')

# Set up the figure for 5 subplots
fig, axes = plt.subplots(3, 2, figsize=(12, 12))

# write the code.
# Plot 1: Count of passengers by class
axes[0,0].bar(df['Pclass'].value_counts().index, df['Pclass'].value_counts(), color='skyblue')
axes[0,0].set_title("Passenger Class Distribution")
axes[0,0].set_xlabel("Pclass")
axes[0,0].set_ylabel("Count")

# Plot 2: Gender distribution
axes[0,1].pie(df['Gender'].value_counts(), labels=df['Gender'].value_counts().index, autopct='%1.1f%%', colors= ['lightblue', 'lightcoral'])
axes[0,1].set_title("Gender Distribution")

# Plot 3: Age distribution
axes[1,0].hist(df['Age'].dropna(), bins=8, color='lightgreen', edgecolor='black')
axes[1,0].set_title("Age Distribution")
axes[1,0].set_xlabel("Age")
axes[1,0].set_ylabel("Frequency")
```

```
# Plot 2: Gender distribution
axes[0,1].pie(df['Gender'].value_counts(), labels=df['Gender'].value_counts().index, autopct='%1.1f%%', colors= ['lightblue', 'lightcoral'])
axes[0,1].set_title("Gender Distribution")

# Plot 3: Age distribution
axes[1,0].hist(df['Age'].dropna(), bins=8, color='lightgreen', edgecolor='black')
axes[1,0].set_title("Age Distribution")
axes[1,0].set_xlabel("Age")
axes[1,0].set_ylabel("Frequency")

# Plot 4: Survival count
axes[1,1].bar(df['Survived'].value_counts().index, df['Survived'].value_counts(), color= ['lightblue', 'lightcoral'])
axes[1,1].set_title("Survival Count")
axes[1,1].set_xlabel("Survived (0 = No, 1 = Yes)")
axes[1,1].set_ylabel("Count")

# Plot 5: Fare vs Age
axes[2,0].scatter(df['Age'], df['Fare'], color='orange', edgecolors='black')
axes[2,0].set_title("Fare vs Age")
axes[2,0].set_xlabel("Age")
axes[2,0].set_ylabel("Fare")

plt.tight_layout()
plt.show()
```


Perform a data visualization for those grains using the Matplotlib library. (Use any 5 different graphs with proper title, legends, axis names, etc. to map identified grains)

